

CS787: GENERATIVE ARTIFICIAL INTELLIGENCE INTRODUCTION

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1st semester, 2026-27

Wed 12:00–13:15 at L20, Fri 12:00–13:15 at L19

Prerequisites

- Computer science
 - Data structures and algorithms
 - Programming, etc.
- Linear algebra
 - Vectors, matrices, etc.
- Calculus
 - Partial derivatives
 - Gradients, Jacobians
 - Maxima, minima, etc.
- Probability and statistics
 - Probabilities
 - Bayes rule
 - Distributions
 - Measures
 - Tests, etc.

General Rules

- Email: `arnabb@cse.iitk.ac.in` and/or `arnabb@iitk.ac.in`
- Put “CS787” in the subject
- Participate
 - Attend classes
 - Clear doubts
 - Answer questions
- Do assignments **individually** unless asked otherwise
- If **you** are sick, follow IITK procedure
 - Take a proper medical leave
 - Produce a sick certificate
- If you are caught **cheating**, appropriate punishment will be given
 - Quantum of punishment decided solely by the instructor
 - No discussion
- General **decorum** should be maintained
 - No laptops or mobiles or tablets or satellite/burner phones, etc.
 - No loud discussions
 - No snoring

Grading Policy

- Exams: 20-30%
- Quiz + Assignments: 10-15%
- Project: 40-50%
 - Form your **own** idea
 - Discuss with instructor and get it approved
 - Just implementation or survey will **not** be enough
 - Back it up with analysis
 - Up to groups of 5
- Paper presentation and discussion: 10-20%
 - **Read** abstracts and shortlist
 - Needs instructor's approval
 - In groups
 - Groups are preferred to be same as projects
- Attendance: 5-10%

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- Changes **may be done** with mutual consent
 - Will be discussed and fixed in the class

- Slides
- Journals/conferences/forums for project and paper ideas, etc.
 - ACL series: ACL, CL, NAACL, EACL, AACL; EMNLP; CoNLL; ...
 - <https://aclanthology.org/>
 - AAAI; KDD; ICML; ...

Course Contents: Part 1

① Artificial Neural Networks

- Feed-forward Networks (FFNN)
- Recurrent Neural Networks (RNN)
- Convolutional Neural Networks (CNN)

② Generative Models

- Hidden Markov Models (HMM)
- Generative Adversarial Networks (GAN), Conditional GAN (cGAN)
- Autoencoders, Variational Autoencoders (VAE)

③ Transformers

- Encoder
- Decoder
- Encoder-decoder

④ (Large) Language Models (LLMs)

- Pre-training
- Continual pre-training
- Fine-tuning

Course Contents: Part 2

- 5 Using LLMs
 - Prompting strategies
 - Thought structures
- 6 Retrieval-augmented Generation (RAG)
- 7 LLM Agents
- 8 Multi-modal LLMs
 - Image
 - Speech
- 9 Privacy and Security of LLMs
 - Memorization
 - Jail-breaking
 - Data stealing
 - Model stealing
 - Attacks
- 10 Applications of Generative AI
- 11 Ethics of Generative AI